

HARSHITH GUJJETI

Tampa, FL | +1 813-786-8772 | harshithgujjeti@icloud.com | [linkedin.com/in/harshithgujjeti](https://www.linkedin.com/in/harshithgujjeti) | github.com/Harshxth | harshxth.github.io

SUMMARY

MS AI & Business Analytics candidate at USF (4.0 GPA) shipping production agentic AI systems. Hackathon-recognized full-stack engineer with deep experience in LangGraph, RAGAS, multi-agent orchestration, and computer vision. Prior AI research at ISRO; applied work across healthcare, voice AI, and real-time CV pipelines.

EDUCATION

University of South Florida — Master of Science, Artificial Intelligence & Business Analytics (GPA: 4.0) *Aug 2025 – May 2027*

Jawaharlal Nehru Technological University — B.Tech, Computer Science (Artificial Intelligence Focus) *Nov 2021 – Jul 2025*

SKILLS

GenAI & Agentic: LangGraph, LangChain, CrewAI, Google ADK, OpenAI API, Gemini, Groq, RAG, RAGAS, MCP, prompt engineering, LLM fine-tuning

NLP, Speech & CV: Whisper, ElevenLabs, spaCy, sentence-transformers, YOLOv8, ByteTrack, SRGAN, CNN, transformers

ML & Data: Python, PyTorch, TensorFlow, scikit-learn, XGBoost, SHAP, pandas, NumPy, SQL, Snowflake

Data Engineering: GCP BigQuery, Airflow, ETL pipelines, medallion architecture (bronze/silver/gold), Power BI, DAX

Infra & MLOps: Docker, FastAPI, Supabase, ChromaDB, FAISS, MongoDB, HuggingFace Spaces, LangSmith, Streamlit

Full-Stack: Next.js, React, TypeScript, Tailwind, Node.js, Vercel, Google Maps API

PROJECT EXPERIENCE

Sensory — Real-Time Accessibility Navigation Platform | [live demo](#) | [github](#) *1st Place — Hackathon UI/UX Track 2026*

- Built full-stack Next.js platform layering noise/crowd/lighting heatmaps on Google Maps with profile-aware route scoring for autistic, wheelchair, deaf, blind, and ESL users across 148 seeded Tampa venues
- Integrated Gemini 2.5 Flash Vision for photo-of-sign wayfinding and ElevenLabs voice cloning (AES-256-GCM encrypted) for multilingual turn-by-turn cues in EN/ES/中文; backed by MongoDB Atlas geospatial + time-series store with OSM wheelchair-access overlays

CareCall — Agentic Voice AI for Post-Discharge Patient Follow-Up | [live demo](#) | [github](#) *Top 5 — HackUSF 2026*

- Built multi-agent system using Google ADK and Gemini API to automate post-discharge patient follow-ups across 70+ languages via ElevenLabs voice synthesis; 9-step care protocol state machine with auto-escalation to nurses on urgent clinical flags
- Deployed AI summarization pipeline delivering structured post-call reports to physicians via email; HIPAA-compliant Supabase backend with encrypted PHI and React dashboard tracking live transcripts, comprehension scoring, and medication confirmation

CitelQ — Agentic RAG System | [live demo](#) | [github](#)

- Built and deployed multi-agent RAG with dynamic LangGraph routing across 3 decision paths and ChromaDB + sentence-transformer retrieval; containerized with Docker and shipped to HuggingFace Spaces as a public demo within a 2-week build cycle
- Achieved 90%+ faithfulness on RAGAS eval across 200+ test queries (0.71 → 0.91 via self-evaluation retry loop); LangSmith tracing on 100% of queries with sub-2s end-to-end latency

Hospital Length-of-Stay Prediction | [github](#)

- Trained XGBoost model on 100,000 patient records using Snowflake-compatible pipelines; achieved MAE of 0.30 days — 5–6× improvement over published benchmarks across 12+ algorithms and 230+ hyperparameter configurations
- Translated SHAP outputs into executive-ready impact narrative: \$1.7–2.6M projected annual savings per 400-bed hospital — mapped technical results to C-suite value

WORK EXPERIENCE

Centre for Urban Transportation Research (CUTR) | *Computer Vision Video Analyst* *Feb 2026 – Present*

- Deployed production YOLOv8 + ByteTrack pipeline processing 100+ hours of traffic video; tracked 1,200+ unique objects across 15+ intersections at 90%+ detection accuracy supporting a federal SS4A safety grant
- Independently scoped and delivered entire pipeline end-to-end; reduced manual annotation time by ~70% and produced 50+ annotated datasets

Muma College of Business, USF | *Research Assistant — Artificial Intelligence* *Sep 2025 – Present*

- Developed and validated ML models for healthcare billing analysis using scikit-learn, XGBoost, and SQL; presented results directly to supervising professor who validated model performance
- Applied HIPAA-aligned data governance across de-identified clinical datasets; contributed to revenue cycle optimization research with measurable reduction in diagnostic coding errors

ISRO — Indian Space Research Organisation | *AI Research Intern* *Jan 2025 – Jul 2025*

- Designed super-resolution pipeline using CNN and SRGAN in TensorFlow under direct supervision of the Head of Analytics; achieved 4× upscaling (10 m → 2.5 m GSD) validated across 1,000+ satellite image pairs
- Presented final methodology and results to 40 scientists with 15+ years of experience in ISRO's Advanced Analytics department — high-stakes technical communication to a senior expert audience

Venkusa Technologies | *AI Developer Intern* *May 2024 – Nov 2024*

- Built AI code-completion tool using OpenAI API with custom prompt chaining and LLM tuning; achieved ~8% efficiency improvement over GitHub Copilot benchmarks across 500+ developer interactions